

Problem 9

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Problem 1 By using linear approximation, determine which of the following is the best estimate of $e^{0.002}$.

(a) 1.00100050016679834166

(b) 1.00200200133400026675

(c) 1.00300450450337702601

(d) 1.02020134002675581016

Explanation. Let $f(x) = e^x$ and $a = 0$. Then since $f'(x) = e^x$, we have that

$$\begin{aligned} L(x) &= f(a) + f'(a)(x - a) \\ &= f(0) + f'(0)(x - 0) \\ &= e^0 + e^0 x \\ &= 1 + x \end{aligned}$$

Then since $L(0.002) = 1 + 0.002 = 1.002$, the answer is (b).